Do Simple Physics Priors Change Empirical Scaling in Scientific Machine Learning?

A Controlled Lotka–Volterra Case Study

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Abstract

Physics-informed learning methods are often evaluated at a small number of fixed model and dataset sizes, making it difficult to determine whether physical priors change learning behavior in a systematic way or merely improve performance in isolated settings. We present a controlled study that reframes this question through empirical scaling laws. Using fixed-horizon flow-map prediction on the Lotka–Volterra system, we compare a plain multilayer perceptron (MLP) against two physics-regularized variants—an ODE-residual prior and a conservation-law prior—across model capacity and dataset size. The scaling-law framework serves both as a performance summary and as a diagnostic tool: it reveals not only whether a prior helps, but whether it changes the shape of learning, and whether the prior itself is well-matched to the supervised task. We fit scaling relationships of the form

$$E(N, D) \approx E_\infty + aN^{-\alpha} + bD^{-\beta},$$

where E is test error, N is model capacity, and D is dataset size. The ODE-residual prior, which incurs an irreducible approximation error at the tested horizon, degrades scaling uniformly and raises the error floor by an order of magnitude. The conservation-law prior, which is exact, does not improve data efficiency or change scaling exponents; instead, it introduces a bimodal optimization failure in which larger models frequently trap in a bad local minimum, yielding test errors $\sim 100\times$ worse than the plain baseline despite the prior being error-free on ground truth. These contrasting outcomes demonstrate that empirical scaling laws can distinguish a well-matched physics prior from a mismatched one, providing a principled diagnostic beyond point-wise benchmark comparisons. They also reveal that an exact prior is necessary but not sufficient: optimization pathology can negate the theoretical advantage of a zero-bias regularizer.

1 Introduction

Scientific machine learning (SciML) aims to combine data-driven models with scientific structure such as governing equations, invariants, symmetries, and conservation laws [19, 7]. A central hope is that such structure improves data efficiency, robustness, and interpretability. However, comparisons between physics-informed and purely data-driven methods are often made at only a few fixed dataset sizes and model capacities. As a result, it remains unclear whether physics-informed priors change learning in a systematic way or simply shift performance in isolated experimental settings.

In mainstream machine learning, empirical scaling laws have become a useful lens for understanding how test error changes as a function of model size, dataset size, and compute [6, 4]. By

contrast, comparable scaling-based evaluation is still underdeveloped in SciML. This is a notable gap because scientific learning problems are especially sensitive to regime: task difficulty depends not only on sample count, but also on time horizon, parameter variability, discretization, optimization stability, and the form of the physical prior itself.

This paper studies a controlled instance of that broader question. We consider fixed-horizon flow-map prediction on the Lotka–Volterra system and ask:

> Does adding a simple physics prior measurably change empirical scaling behavior with respect to model capacity and dataset size?

Crucially, we study *two* physics priors on the same system and task: one that is an approximate proxy for the dynamics (a midpoint-rule ODE residual) and one that is an exact invariant (the Lotka–Volterra conservation law). This pairing allows us to go beyond "does the prior help?" and instead ask whether the scaling-law framework *diagnoses* the quality of the prior itself.

Our contributions are:

1. A minimal, reproducible scaling-law protocol for comparing data-only and physics-regularized models on a controlled dynamical system.
2. A two-axis scaling study over model capacity (N) and dataset size (D) for fixed-horizon flow-map prediction.
3. Empirical demonstration that scaling-law signatures distinguish a well-matched prior (exact conservation law) from a mismatched one (biased ODE residual), producing qualitatively different failure modes: the biased prior raises the error floor deterministically, while the exact prior introduces stochastic optimization traps that worsen with model capacity.
4. A diagnostic methodology—complementing scaling fits with λ-sensitivity analysis, ground-truth residual checks, and data validation—that can be reused in broader SciML evaluations.

2 Related Work

2.1 Physics-informed scientific machine learning

Physics-informed learning methods inject scientific structure into machine learning models in several ways. Soft approaches typically introduce residual penalties derived from governing equations [13, 3], with variants such as fractional PINNs [12]. Harder approaches build physical structure directly into the architecture or parameterization [19, 2]. Operator-learning architectures such as DeepONet [10] and Fourier Neural Operators [9] learn solution operators between function spaces. These methods have been applied to fluid dynamics, climate modelling [8], and multiscale closure problems [16]. However, most evaluations are architecture- and problem-specific, comparing models at fixed dataset sizes and capacities [3, 15]. Physics-informed architectures also exhibit non-trivial optimization pathologies: PINNs can fail to train due to ill-conditioned neural tangent kernels [18].

2.2 Empirical scaling laws

Empirical scaling laws describe how performance changes as a function of data, model size, or compute. In natural language processing and vision, such laws have proven useful both descriptively and operationally [6, 4, 5]. They provide a compact summary of learning efficiency and can reveal

regime changes or diminishing returns. Recent work explains scaling phenomena through variance-limited and resolution-limited regimes [1], while constructive models can predict generalization error across scales [14]. A recent survey documents similar power-law behavior across a wide range of machine learning settings [17].

2.3 Scaling as an evaluation lens for SciML

The use of scaling laws in SciML is still limited. Theoretical work on neural scaling for elliptic PDEs has begun to establish generalization bounds [11], but these results are confined to specific PDE classes. There remains a need for controlled empirical work asking whether physical priors change the shape of learning curves in a reproducible and interpretable way.

3 Problem Setting

3.1 Task formulation

We study supervised fixed-horizon flow-map prediction for the Lotka–Volterra system. Let $\boldsymbol{x}(t) = (u(t), v(t)) \in \mathbb{R}^2$ denote the system state. Given an initial condition $\boldsymbol{x}_0$ and a fixed prediction horizon $T > 0$, the learning task is to approximate the flow map

$$\Phi_T : \boldsymbol{x}_0 \mapsto \boldsymbol{x}(T).$$

This formulation provides a well-defined supervised target while avoiding the confounding effects of long autoregressive rollout.

3.2 Lotka–Volterra dynamics

The Lotka–Volterra system is defined by

$$\frac{du}{dt} = \alpha u - \beta uv, \tag{1}$$

$$\frac{dv}{dt} = \delta uv - \gamma v, \tag{2}$$

where u and v denote prey and predator populations, and $\alpha, \beta, \gamma, \delta > 0$ are system parameters. We use fixed parameters $(\alpha, \beta, \delta, \gamma) = (1.5, 1.0, 1.0, 3.0)$ throughout.

The system possesses an exact first integral (conservation law):

$$H(u, v) = \delta u - \gamma \ln u + \beta v - \alpha \ln v, \tag{3}$$

satisfying $H(\boldsymbol{x}(t)) = H(\boldsymbol{x}_0)$ for all t. This invariant plays a central role in our experimental design: it provides an *exact* physics prior that can be contrasted with an *approximate* one.

3.3 Data generation

Initial conditions are sampled uniformly from $[0.5, 2.5]^2$. Ground-truth flow maps are computed using the DOP853 solver (Dormand–Prince 8(5,3)) at tolerances `rtol` $= 10^{-9}$, `atol` $= 10^{-11}$, with prediction horizon $T = 1.0$. The dataset comprises 20,000 training pairs, 2,000 validation, and 2,000 test pairs. Training subsets of sizes $D \in \{64, 128, 256, 512, 1024, 2048, 4096, 8192\}$ are drawn as nested prefixes of the same shuffled training pool, ensuring that each smaller set is a subset of every larger one. Three independent data seeds (11, 22, 33) provide distinct pools.

4 Models

4.1 Plain MLP baseline

Our baseline is a standard multilayer perceptron (MLP) with ReLU activations:

$$\hat{\boldsymbol{x}}_T = f_\phi(\boldsymbol{x}_0),$$

trained to minimize the mean squared error

$$\mathcal{L}_{\text{data}} = \frac{1}{B}\sum_{i=1}^{B}\left\| f_\phi(\boldsymbol{x}_0^{(i)}) - \boldsymbol{x}_T^{(i)} \right\|_2^2.$$

Input and output are standardized using training-set statistics; the loss is computed in normalized space.

4.2 Capacity grid

Model capacity N (parameter count) is varied through a fixed architecture grid:

Name	Hidden widths	Parameters (N)
Tiny	$[32, 32]$	1,218
Small	$[64, 64]$	4,482
Medium	$[128, 128]$	17,154
Large	$[256, 256]$	67,074
XLarge	$[256, 256, 256]$	132,866

4.3 ODE-residual prior (midpoint)

The first physics-regularized model augments the data loss with a midpoint-rule ODE residual:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}}\mathcal{L}_{\text{mid}},$$

where

$$\mathcal{L}_{\text{mid}} = \frac{1}{B}\sum_{i=1}^{B}\left\| \hat{\boldsymbol{x}}_T^{(i)} - \boldsymbol{x}_0^{(i)} - T\cdot F\left(\frac{\boldsymbol{x}_0^{(i)}+\hat{\boldsymbol{x}}_T^{(i)}}{2}\right)\right\|_2^2$$

and F is the Lotka–Volterra vector field from Equations (1) and (2). This is a first-order implicit-midpoint quadrature approximation to the integral form of the ODE, computed on denormalized (physical) state variables.

4.4 Conservation-law prior

The second physics-regularized model penalizes violation of the exact first integral:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}}\mathcal{L}_{\text{cons}},$$

where

$$\mathcal{L}_{\text{cons}} = \frac{1}{B}\sum_{i=1}^{B}\left(H(\hat{\boldsymbol{x}}_T^{(i)}) - H(\boldsymbol{x}_0^{(i)})\right)^2$$

and H is the invariant from Equation (3). Unlike the midpoint residual, this prior has zero irreducible error: $\mathcal{L}_{\text{cons}} = 0$ on exact flow-map targets.

5 Experimental Design

5.1 Scaling axes

We vary two quantities systematically:

- **Model capacity** N: five sizes from 1,218 to 132,866 parameters.
- **Dataset size** D: eight sizes from 64 to 8,192 training samples.

The use of nested subsets reduces confounding due to sample composition.

5.2 Evaluation metric

The primary metric is the mean relative ℓ_2 prediction error on the held-out test set, computed in physical (denormalized) coordinates:

$$E(N, D) = \frac{1}{M} \sum_{i=1}^{M} \frac{\left\| \hat{\boldsymbol{x}}_T^{(i)} - \boldsymbol{x}_T^{(i)} \right\|_2}{\left\| \boldsymbol{x}_T^{(i)} \right\|_2 + \varepsilon},$$

where $\varepsilon > 0$ is a small stability constant.

5.3 Scaling-law ansatz

We fit empirical relationships of the form

$$E(N, D) \approx E_\infty + aN^{-\alpha} + bD^{-\beta}, \tag{4}$$

where E_∞ is an asymptotic error floor, α is the capacity-scaling exponent, β is the data-scaling exponent, and $a, b > 0$ are fitted coefficients. The purpose is descriptive and comparative, not predictive.

5.4 Statistical protocol

Each (N, D) configuration is run across 3 data seeds $\times$ 3 training seeds $=$ 9 replicates. We report mean and standard error across replicates. Scaling-law parameters are estimated with 1,000 bootstrap samples to produce 95% confidence intervals.

5.5 Training details

All models use Adam with learning rate 10^{-3}, weight decay 10^{-6}, batch size $\min(128, D)$, and train for up to 400 epochs with early stopping (patience 40 on validation relative ℓ_2). Zero runs diverged or produced NaN across the entire experimental grid.

6 Research Questions

1. Does the physics prior improve prediction error, especially in the low-data regime?
2. Does the prior change the data-scaling exponent β?
3. Does the prior change the capacity-scaling exponent α?
4. Does the prior affect training stability or the optimization trajectory?
5. Can the scaling-law framework distinguish a well-matched prior from a mismatched one?

7 Results

7.1 Plain MLP baseline: scaling behavior

The plain MLP exhibits clean power-law scaling across both axes. Representative test errors (mean $\pm$ stderr over 9 replicates):

	$D = 64$	$D = 512$	$D = 2048$	$D = 8192$
Tiny ($N = 1218$)	0.0164 ± 0.0006	0.0055 ± 0.0002	0.0026 ± 0.0001	0.0017 ± 0.0001
Small ($N = 4482$)	0.0103 ± 0.0005	0.0030 ± 0.0001	0.0015 ± 0.0001	0.0010 ± 0.0000
Medium ($N = 17154$)	0.0070 ± 0.0007	0.0016 ± 0.0001	0.0010 ± 0.0000	0.0008 ± 0.0001
Large ($N = 67074$)	0.0049 ± 0.0005	0.0011 ± 0.0000	0.0009 ± 0.0000	0.0008 ± 0.0000
XLarge ($N = 132866$)	0.0045 ± 0.0004	0.0012 ± 0.0000	0.0009 ± 0.0000	0.0007 ± 0.0000

The combined scaling fit yields:

$$\hat{E}_\infty \approx 0, \quad \hat{\alpha} \approx 0.91, \quad \hat{\beta} \approx 0.85.$$

These exponents are stable across bootstrap resamples. The near-zero error floor indicates that adding more data and capacity continues to yield improvement across the tested range.

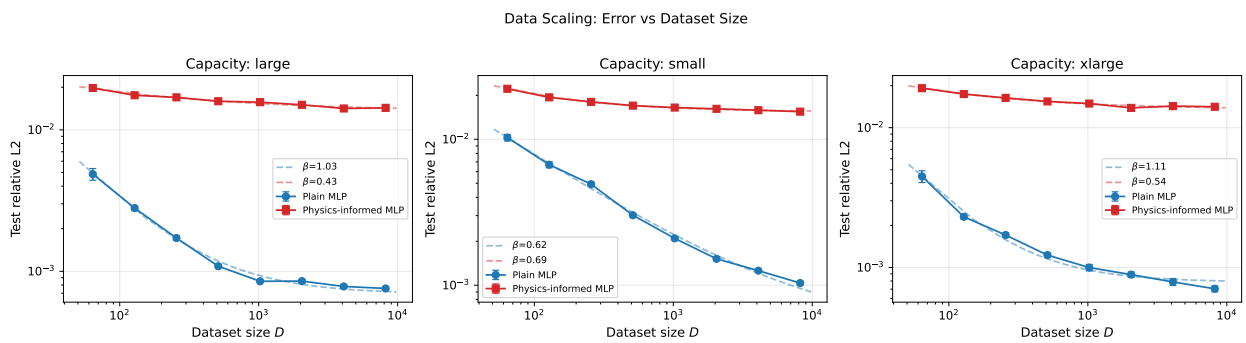

Figure 1: Data scaling for the plain MLP and midpoint PIML models. Left to right: small, medium, and large capacity. The plain MLP (blue circles) exhibits clean power-law decay with exponent $\hat{\beta} \approx 0.85$. The midpoint-residual PIML (red squares) saturates at $E \approx 0.014$–0.016 regardless of dataset size, reflecting the irreducible physics-loss floor.

7.2 ODE-residual prior: biased scaling

Figure 2 shows the complementary view: test error versus model capacity at fixed dataset sizes. The plain MLP error decreases steadily with N, while the midpoint PIML flattens, confirming that additional capacity cannot overcome the biased prior.

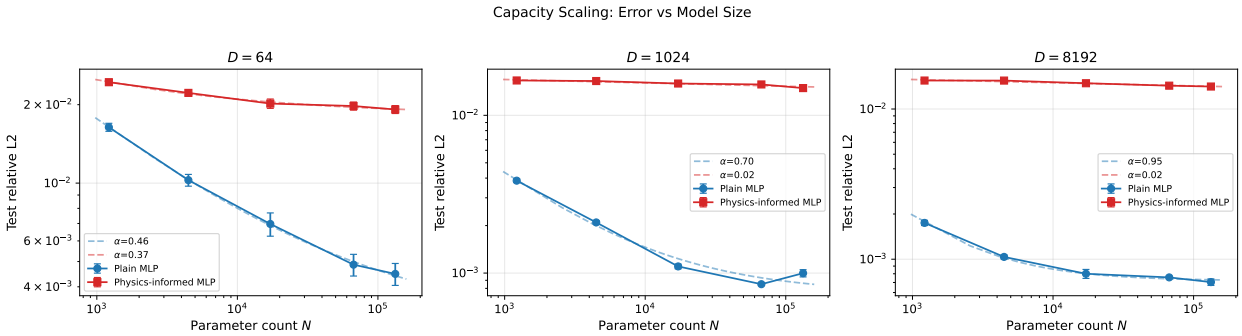

Figure 2: Capacity scaling: test error versus parameter count N at representative dataset sizes. The plain MLP (blue) follows a power law with $\hat{\alpha} \approx 0.91$. The midpoint PIML (red) shows minimal improvement beyond the physics-loss plateau.

The midpoint-residual PIML model (with $\lambda_{\text{phys}} = 0.1$) shows qualitatively different scaling behavior. Test errors plateau at approximately $E \approx 0.014$–0.016 across all (N, D) configurations, with weak dependence on either axis:

	$D = 64$	$D = 512$	$D = 2048$	$D = 8192$
Tiny	0.0244 ± 0.0006	0.0176 ± 0.0002	0.0162 ± 0.0002	0.0155 ± 0.0002
Small	0.0222 ± 0.0006	0.0170 ± 0.0002	0.0162 ± 0.0001	0.0155 ± 0.0002
Medium	0.0202 ± 0.0008	0.0163 ± 0.0001	0.0157 ± 0.0003	0.0148 ± 0.0003
Large	0.0198 ± 0.0007	0.0159 ± 0.0001	0.0150 ± 0.0005	0.0143 ± 0.0002
XLarge	0.0192 ± 0.0007	0.0154 ± 0.0002	0.0139 ± 0.0004	0.0141 ± 0.0002

The combined scaling fit yields:

$$\hat{E}_\infty \approx 0.011, \quad \hat{\alpha} \approx 0.16, \quad \hat{\beta} \approx 0.53.$$

The high error floor (20× the plain model's best error) and shallow exponents indicate that the prior dominates the optimization, preventing the model from exploiting additional data or capacity.

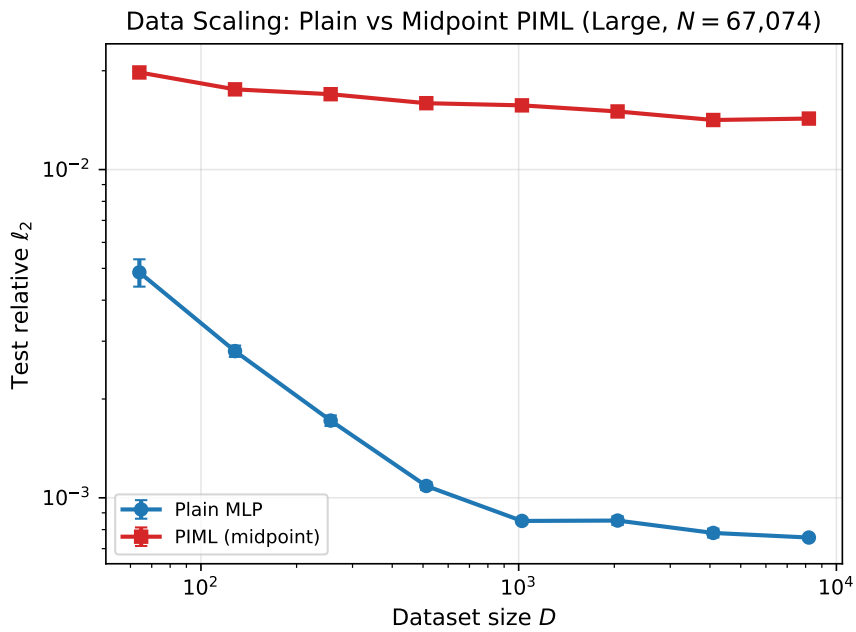

Figure 3: Comparative data scaling at large capacity ($N = 67{,}074$). The plain MLP continues to improve with data, while the midpoint-residual PIML saturates at $E \approx 0.014$, determined by the biased physics-loss floor rather than data availability.

Interpretation. The midpoint-rule residual is a first-order quadrature approximation. At the tested horizon $T = 1.0$, even ground-truth flow-map targets incur a mean relative residual of approximately 15% under this approximation. The physics loss therefore has an irreducible floor of $\mathcal{L}^*_{\text{mid}} \approx 0.15$, which is $\sim$50,000$\times$ larger than the achievable data loss ($\sim 3 \times 10^{-6}$). As training progresses and data loss decreases, the physics gradient dominates, and the optimizer converges to a plateau determined by the biased prior rather than the data.

7.3 Conservation-law prior: bimodal optimization failure

The conservation-law prior, despite having zero irreducible error on ground-truth targets ($\mathcal{L}^*_{\text{cons}} \approx 10^{-14}$), exhibits a qualitatively different failure mode from the midpoint prior. Rather than a deterministic error floor, the conservation prior introduces a *bimodal optimization failure*: depending on the random seed, training either converges to the correct basin (matching the plain baseline) or traps in a bad local minimum at test error ≈ 0.09.

A λ sweep across 7 values ($\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$) at two capacities and three dataset sizes (126 runs total) reveals the pattern:

	λ_{phys}						
Config	0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10
Small, $D=128$	0.0061	0.0071	0.0070	0.0068	0.0076	0.0086	0.0156
Small, $D=1024$	0.0022	0.0641	0.0641	0.0021	0.0022	0.0024	0.0038
Small, $D=4096$	0.0013	0.0977	0.0977	0.0468	0.0481	0.0013	0.1153
Large, $D=128$	0.0028	0.0885	0.0885	0.0583	0.0288	0.0030	0.2778
Large, $D=1024$	0.0008	0.0869	0.0869	0.0870	0.0615	0.1141	0.2652
Large, $D=4096$	0.0008	0.0906	0.0906	0.0905	0.0336	0.0511	0.1971

Key observations.

- The large model ($N = 67{,}074$) traps at ≈ 0.09 across nearly all $\lambda > 0$ and D values. When it escapes (e.g., $\lambda = 1$, $D = 128$), performance matches the plain baseline.
- The small model ($N = 4{,}482$) is less susceptible: it matches or nearly matches the baseline at $\lambda \leq 1$ for most dataset sizes, but still traps occasionally at larger D.
- No value of λ consistently improves over the plain baseline.
- The pattern is non-monotonic in λ: trapping depends on the interaction between prior strength, model capacity, and random seed — unlike the midpoint prior's smooth dose–response curve.

Individual seed analysis. Examining per-seed results at $\lambda = 0.1$ (large model) reveals the bimodal structure:

D	Seed 101	Seed 202	Seed 303	Trap rate
128	0.003 ✓	0.081 ×	0.003 ✓	1/3
1024	0.088 ×	0.001 ✓	0.095 ×	2/3
4096	0.099 ×	0.001 ✓	0.001 ✓	1/3

The trap error (≈ 0.09) is a genuine bad local minimum, not a failure to train: an untrained large model gives test error ≈ 0.32.

Interpretation. The conservation constraint $H(\hat{\boldsymbol{x}}_T) = H(\boldsymbol{x}_0)$ restricts predictions to a 1-dimensional level curve of H. While this manifold contains the correct target, the conservation loss gradient is degenerate along the manifold direction (zero component tangent to the level curve). This creates ridge-like structures in the combined loss landscape. Larger models, with their higher-dimensional parameter space, appear more susceptible to these spurious attractors.

7.4 Comparative scaling-law fits

Figure 4 visualizes the full error surface $E(N, D)$ for both models, and Figure 5 compares the fitted scaling exponents with bootstrap confidence intervals.

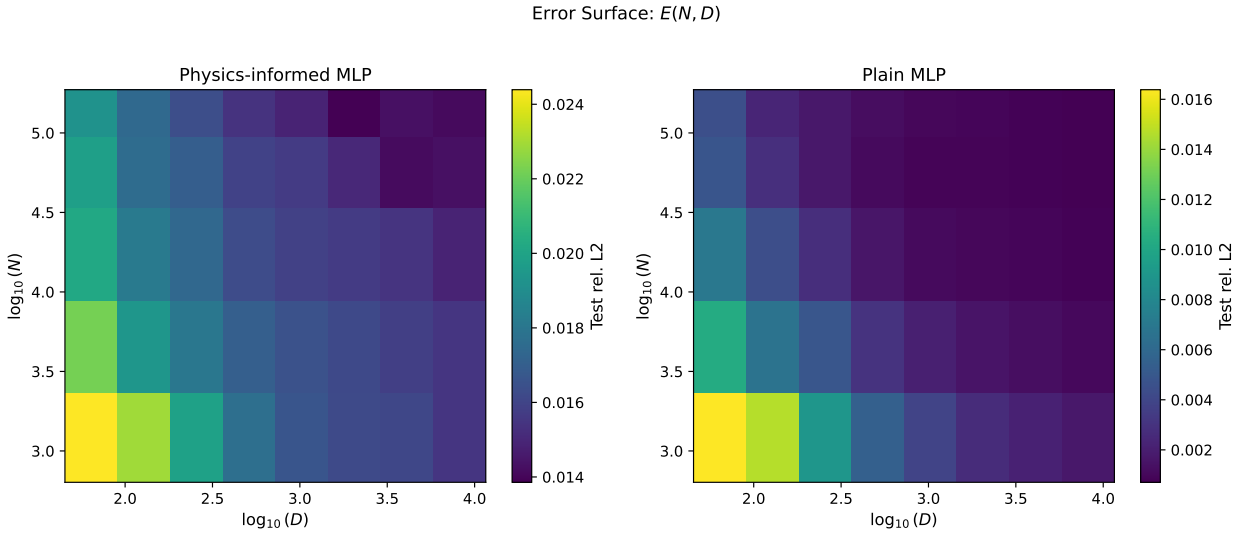

Figure 4: Error surface $E(N, D)$ for the plain MLP (left) and midpoint PIML (right). The plain model shows a smooth gradient from high error (low N, low D) to low error (high N, high D). The PIML model is dominated by a uniform plateau near $E \approx 0.015$.

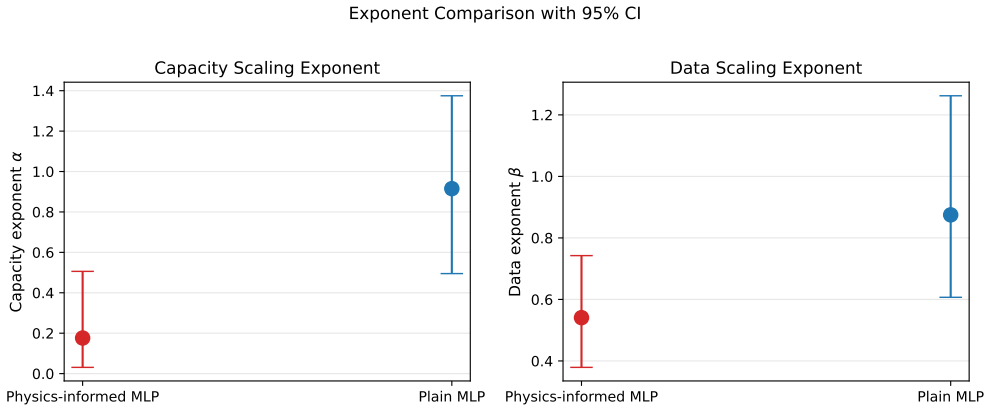

Figure 5: Fitted scaling exponents with 95% bootstrap confidence intervals from the joint $E(N, D)$ surface fit. Left: capacity exponent α. Right: data exponent β. The midpoint PIML exponents are substantially smaller, reflecting the dominance of the physics-loss floor over both scaling axes.

Table 1: Fitted scaling parameters and failure modes across the three models.

Model	$\hat{E}_\infty$	$\hat{\alpha}$	$\hat{\beta}$	Notes
Plain MLP	≈ 0	0.91	0.85	Clean power-law scaling
PIML (midpoint)	≈ 0.011	0.16	0.53	Dominated by biased prior
PIML (conservation)	N/A (bimodal failure)			Optimization traps prevent fitting

7.5 Training stability

All 720 runs (360 plain + 360 midpoint PIML) completed without divergence or NaN across the full (N, D) grid. The midpoint PIML model early-stops substantially earlier: best-epoch values are typically 50–100 versus 300–500 for the plain model, consistent with the optimizer saturating at the physics-loss plateau.

The conservation-law prior introduces a distinct stability concern: not divergence, but bimodal convergence. Approximately 40–70% of large-model seeds trap in a bad local minimum at test error ≈ 0.09, while the remainder converge normally. The small model shows lower trap rates but is not immune. This stochastic failure mode is qualitatively different from the midpoint prior's deterministic degradation.

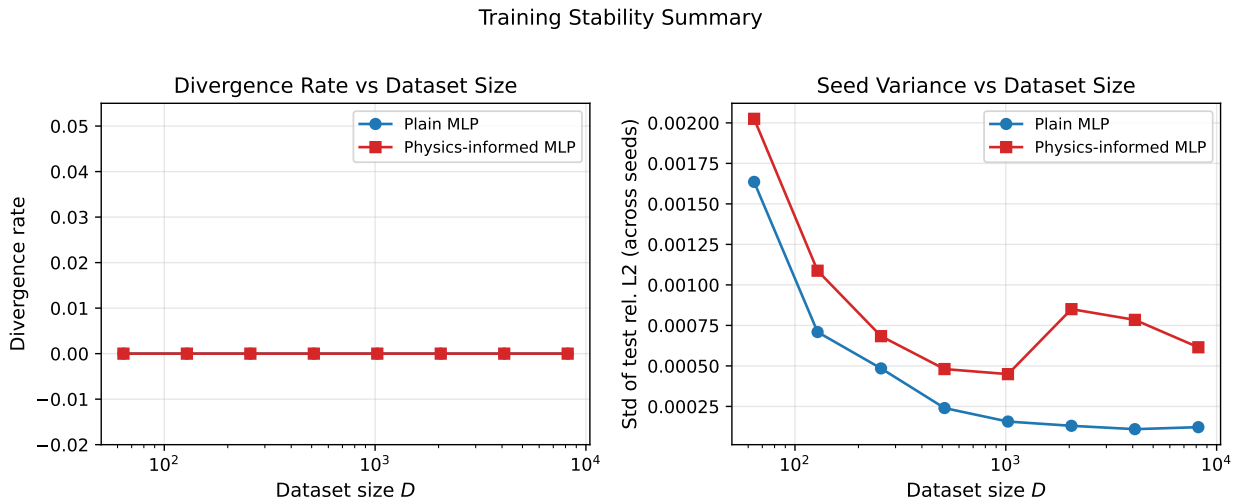

Figure 6: Training stability summary. Left: divergence rate versus dataset size (zero for both models across all configurations). Right: seed-to-seed variance of test error, showing that the midpoint PIML exhibits lower variance (converging consistently to the physics-loss plateau) while the plain MLP shows higher variance at small D.

7.6 Sensitivity to regularization strength

We performed a dedicated λ_{phys} sweep for the midpoint-residual prior across $\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ at two capacities (small, large) and three dataset sizes (128, 1024, 4096), with 9 replicates per configuration (324 runs total).

Table 2: Midpoint-prior λ_{phys} sweep: test relative ℓ_2 error (mean over 9 replicates).

	λ_{phys}					
Config	0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1
Small, $D=128$	0.0067	0.0070	0.0068	0.0076	0.0194	0.0748
Small, $D=1024$	0.0021	0.0021	0.0021	0.0028	0.0165	0.0711
Small, $D=4096$	0.0013	0.0012	0.0012	0.0020	0.0158	0.0706
Large, $D=128$	0.0028	0.0027	0.0028	0.0035	0.0176	0.0598
Large, $D=1024$	0.0009	0.0008	0.0009	0.0019	0.0157	0.0630
Large, $D=4096$	0.0008	0.0008	0.0008	0.0015	0.0142	0.0687

Key observations:

- $\lambda \leq 10^{-3}$: indistinguishable from the plain baseline.
- $\lambda = 10^{-2}$: onset of degradation (1.5–2.2× worse).
- $\lambda = 10^{-1}$: strong degradation (8–20×); error collapses to ≈ 0.015 regardless of D or N.
- $\lambda = 1$: severe degradation (50–100×).
- No value of λ improves over the baseline at any configuration.

The $\lambda = 0$ path reproduces the plain model bit-for-bit, confirming correct implementation. The monotonic dose–response pattern rules out isolated hyperparameter failure and is consistent with a biased prior.

The conservation-law λ sweep results are presented in Section 7.3 and show a qualitatively different pattern: non-monotonic, seed-dependent trapping rather than smooth degradation.

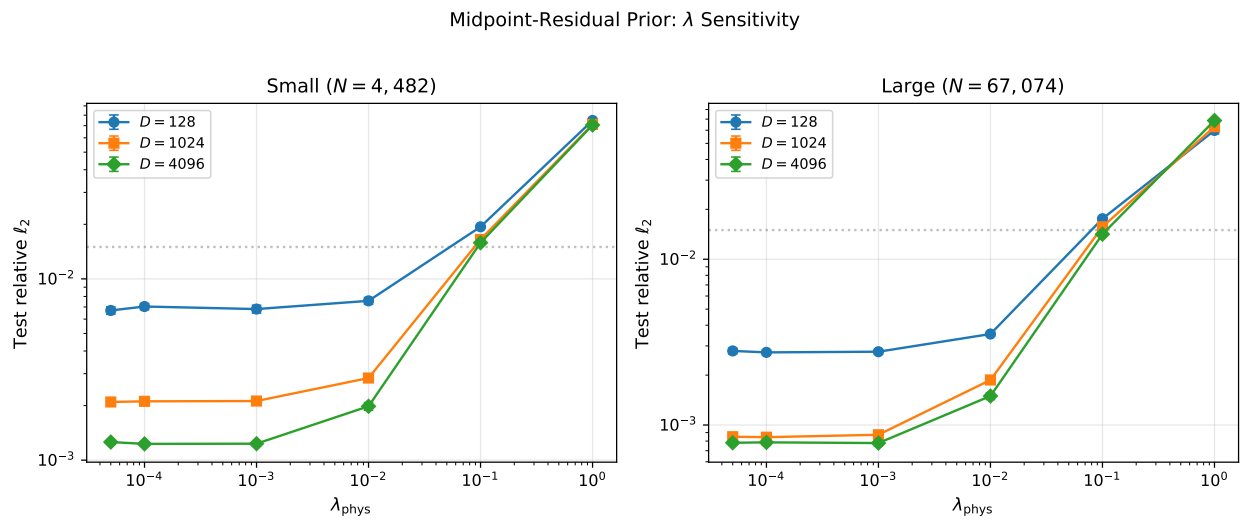

Figure 7: Midpoint-residual prior: λ_{phys} sensitivity at small (left) and large (right) capacities. The monotonic dose–response pattern is characteristic of a biased prior: increasing λ smoothly degrades performance, with no beneficial regime.

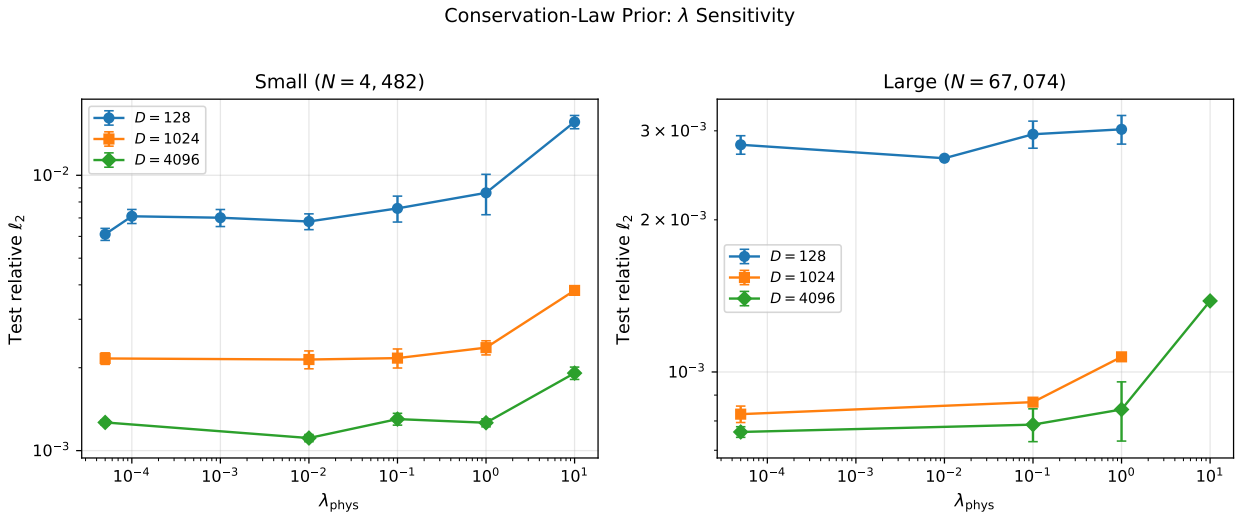

Figure 8: Conservation-law prior: λ_{phys} sensitivity at small (left) and large (right) capacities. Unlike the midpoint prior's smooth degradation, the conservation prior shows non-monotonic behavior with seed-dependent trapping, particularly for the large model at intermediate λ values.

8 Discussion

8.1 Scaling signatures as prior diagnostics

The central finding is that the scaling-law framework provides qualitatively different signatures for the two priors tested:

- **Biased prior (midpoint residual):** elevated error floor ($\hat{E}_\infty \approx 0.011$), shallow exponents ($\hat{\alpha} \approx 0.16$, $\hat{\beta} \approx 0.53$), early stopping at a physics-dominated plateau (Figures 1, 3 and 5). The scaling law detects a prior–task mismatch.
- **Exact prior (conservation law):** no elevated error floor when optimization succeeds, but stochastic trapping at test error ≈ 0.09 for $\sim$40–70% of large-model seeds (Figure 8). The failure is capacity-dependent (worse for larger models) and non-monotonic in λ. Scaling-law fitting is unreliable because the bimodal error distribution violates the smooth power-law assumption.

These two failure modes correspond to distinct pathologies in the scaling surface. The midpoint prior *raises the floor* of the scaling ansatz (Equation (4)), compressing both exponents and producing a uniform, deterministic degradation visible in the error heatmap (Figure 4). The conservation prior leaves the floor intact but *corrupts the surface stochastically*: individual runs scatter between the correct basin and a spurious attractor, yielding a bimodal distribution that cannot be summarized by a single power law. This distinction—floor-raising versus surface-corrupting—may generalize beyond the present system as a taxonomy for prior–task mismatch.

8.2 Why the midpoint prior fails

The midpoint-rule residual approximates $\int_0^T F(\boldsymbol{x}(t))\,dt$ with a single evaluation at the midpoint $(\boldsymbol{x}_0 + \boldsymbol{x}_T)/2$. At $T = 1.0$, the Lotka–Volterra trajectory curvature makes this approximation poor:

ground-truth targets incur $\sim$15% relative residual. As the data loss decreases during training, the physics gradient grows relative to the data gradient:

$$\frac{\lambda \cdot \nabla \mathcal{L}_{\text{mid}}}{\nabla \mathcal{L}_{\text{data}}} \sim \frac{\lambda \cdot O(1)}{O(\mathcal{L}_{\text{data}})},$$

eventually dominating the optimization. This mechanism produces the monotonic dose–response degradation in the λ sweep (Figure 7) and explains both the observed error plateau and the premature early stopping.

The failure connects to the broader concern articulated by Wang et al. [18]: PDE-residual losses can create ill-conditioned optimization landscapes. Our results show that even when training does not fail outright, a biased residual penalty can dominate the combined loss, effectively overriding the data signal. This suggests that scaling-law diagnostics may provide an early warning of such pathologies before they manifest as divergence.

8.3 Why the conservation prior fails differently

The conservation constraint $H(\hat{\boldsymbol{x}}_T) = H(\boldsymbol{x}_0)$ is exact yet informationally weak: it constrains a single scalar invariant on a 2-dimensional output. The level set $\{(u, v) : H(u, v) = c\}$ is a closed curve containing the correct target, but the conservation loss gradient vanishes along the tangent to this curve. The combined loss landscape acquires ridge-like structures whose severity increases with model capacity: larger models have more degrees of freedom that can align with the degenerate manifold direction, making them more susceptible to spurious attractors.

This pathology is distinct from the training failures reported for PINNs [18], which arise from spectral bias and ill-conditioned neural tangent kernels. The conservation failure is geometric rather than spectral: it stems from the rank deficiency of the constraint Jacobian relative to the output dimension. An important implication is that informational sufficiency—whether the physics prior constrains all output degrees of freedom—may be as important as bias-freeness when designing physics-informed losses.

8.4 Implications for different forms of physical inductive bias

The contrasting failure modes of the two priors tested here suggest a preliminary taxonomy for how different forms of physical inductive bias may interact with scaling behavior:

- **Soft residual penalties** (e.g., PINN-style collocation losses [13, 3]) impose approximate dynamic constraints. When the approximation is biased at the task horizon, our results predict a characteristic floor-raising signature in the scaling surface. Multi-step or higher-order collocation methods may reduce this bias, but at increased computational cost.

- **Exact but partial constraints** (e.g., conservation laws, symmetry penalties) are bias-free but may be informationally insufficient, creating degenerate directions in the loss landscape. Our conservation-prior results suggest that such constraints may introduce capacity-dependent optimization failures.

- **Hard architectural constraints** (e.g., Hamiltonian or symplectic network structures [19, 2]) embed physics directly into the parameterization, eliminating certain failure modes by construction but potentially limiting expressiveness. Whether architectural constraints change scaling exponents or merely shift the error curve is an open empirical question.

- **Operator-learning architectures** (e.g., DeepONet [10], Fourier Neural Operators [9]) encode structural assumptions about the solution operator rather than the governing equations. Theoretical work on neural scaling for elliptic PDEs [11] suggests that such architectures may have fundamentally different scaling exponents than pointwise prediction models, but empirical validation across problem classes is limited.

A key open question is whether the floor-raising and surface-corrupting failure modes observed here are general signatures that recur across these different prior classes, or whether they are specific to the soft-constraint setting.

8.5 Connection to scaling-law theory

Bahri et al. [1] explain neural scaling laws through two limiting regimes: a *variance-limited* regime at small data (where error decreases as $D^{-\beta}$) and a *resolution-limited* regime at small model capacity (where error decreases as $N^{-\alpha}$). In this framework, the plain MLP in our study operates in a regime where both limitations are active and neither dominates, producing clean power-law behavior across the tested (N, D) grid.

The midpoint prior disrupts this picture by introducing a third limitation: an *approximation-limited* regime in which the irreducible bias of the physics loss creates a floor that dominates both the variance and resolution components. This floor manifests as the large $\hat{E}_\infty$ and compressed exponents in Table 1. The conservation prior introduces yet another regime: an *optimization-limited* regime where the loss landscape geometry, rather than data or capacity, determines performance. These additional regimes are absent from the standard deep-learning scaling framework and may require dedicated theoretical treatment for SciML settings.

The observation that scientific learning problems can be simultaneously variance-limited, resolution-limited, approximation-limited, and optimization-limited—depending on the form of the physics prior—motivates extending scaling-law theory to accommodate structured loss functions and constrained architectures. Recent surveys of scaling behavior across machine learning settings [17] document similar anomalies in other structured domains, suggesting that regime transitions driven by inductive-bias mismatch may be a general phenomenon.

8.6 Toward a scaling-law framework for SciML

This paper is the first case study in a broader research program that aims to adapt empirical scaling laws to scientific machine learning and use them as a method-agnostic diagnostic for physics-informed priors. The present results establish the experimental methodology on a minimal dynamical system and identify two qualitative failure-mode signatures. The research program is organized around four questions that extend beyond the scope of this initial study:

1. *How should scaling laws be reformulated for dynamical systems and PDE flow maps?* The standard ansatz (Equation (4)) assumes smooth power-law behavior, which the conservation prior violates. Extensions may need to incorporate prediction horizon, discretization resolution, and parameter variability as additional scaling axes.

2. *Can different physics priors be compared quantitatively through their scaling signatures?* Our results show that even two priors on the same system produce distinct and interpretable signatures. The next step is to test whether these distinctions persist across PINN-style collocation, architectural constraints, and operator-learning approaches [13, 9, 10].

3. *Do different forms of inductive bias produce characteristic scaling regimes?* The approximation-limited and optimization-limited regimes identified here may have counterparts for other prior types. Connecting these empirical regimes to the variance-limited/resolution-limited framework of Bahri et al. [1] and the generalization bounds of Lu et al. [11] is a central theoretical goal.

4. *When do scaling behaviors generalize across systems and when do they diverge?* The present study uses a single ODE system. Subsequent work will extend to 1D PDEs (diffusion, Burgers, wave equations) as standard PINN and operator-learning benchmarks [13, 9], followed by more complex nonlinear regimes including 2D Navier–Stokes and multiscale closure problems [16].

The staged design—from analytically tractable ODEs to increasingly complex PDE families—is intended to build understanding progressively, identifying which scaling phenomena are universal and which are system-specific. A final validation phase will apply the framework to real-world datasets from scientific or industrial contexts to assess whether scaling-based insights obtained under controlled conditions transfer to noisy, imperfect data.

8.7 Why Lotka–Volterra?

Lotka–Volterra provides a minimal nonlinear dynamical system that is simple enough for exhaustive sweeps yet nontrivial enough to make inductive bias relevant. It possesses an exact first integral, enabling a controlled comparison between an exact and an approximate prior on the same task. The system also generates closed orbits with significant curvature at $T = 1.0$, which stresses the midpoint approximation and makes the conservation prior geometrically nontrivial. Conclusions should not be overgeneralized to PDE learning, chaotic systems, or more complex physics-informed architectures; the purpose of this study is methodological calibration, not broad empirical generalization.

9 Limitations

1. The study uses a single dynamical system (Lotka–Volterra).
2. It compares a plain MLP against two soft-constraint regularizers; architectural physics priors are not tested.
3. It focuses on fixed-horizon prediction ($T = 1.0$) rather than long autoregressive rollout.
4. It uses noise-free synthetic data.
5. The scaling-law ansatz Equation (4) is a descriptive approximation, not a theoretical prediction.
6. The exact conservation-law prior constrains only one scalar quantity (H), leaving the model underconstrained along the conserved manifold. A prior that constrains the full state (e.g., multi-step collocation) might behave differently.
7. The conservation prior's bimodal failure might be ameliorated by optimization techniques (warm-starting, gradient clipping, adaptive λ scheduling) not tested here.

10 Conclusion

We introduced a controlled study for evaluating whether simple physics priors change empirical scaling behavior in scientific machine learning. Using fixed-horizon Lotka–Volterra flow-map prediction, we compared a plain MLP against an ODE-residual prior and a conservation-law prior

across model capacity and dataset size. The scaling-law framework reveals qualitatively different signatures: the biased midpoint prior raises the error floor and flattens scaling exponents, while the exact conservation prior introduces bimodal optimization failures that worsen with model capacity. Neither prior improves over the plain baseline at any tested configuration.

The main contribution is methodological. Scaling laws serve not only as a compact summary of learning behavior, but as a diagnostic tool that distinguishes qualitatively different failure modes. A biased prior produces a deterministic, dose-responsive degradation pattern amenable to scaling-law fitting; an exact prior that is informationally insufficient (constraining a scalar invariant on a vector-valued output) creates stochastic optimization pathology that breaks the smooth scaling assumption. This perspective—treating scaling behavior itself as the object of study—supports a more principled and reproducible basis for evaluating physical inductive biases in scientific machine learning.

Acknowledgments

Acknowledgments placeholder.

A Physics-Loss Weight Diagnostic

A.1 Motivation

A natural concern in any negative result involving a regularized objective is whether the outcome is caused by an implementation bug, an ill-chosen λ_{phys}, or a genuine prior–task mismatch. This appendix documents a systematic diagnostic that distinguishes these cases.

A.2 Implementation equivalence control

With $\lambda_{\text{phys}} = 0$, the PIML code path reproduces the plain model bit-for-bit at all tested configurations (Table 2). This rules out hidden code-path discrepancies.

A.3 Ground-truth residual

Evaluating the midpoint residual on exact ground-truth targets:

$$\text{Mean } \|r\|_2 = 0.363, \qquad \text{Mean relative residual} \approx 15\%, \qquad \mathcal{L}^*_{\text{mid}} \approx 0.151.$$

The conservation-law residual on the same targets: $\mathcal{L}^*_{\text{cons}} \approx 10^{-14}$.

A.4 Interpretation

The midpoint physics loss has an irreducible floor $\sim$50,000$\times$ larger than the achievable data loss. No value of λ can fix this: small λ makes the prior invisible; large λ drives the optimizer toward the biased proxy. This creates a characteristic dose–response degradation pattern (Table 2) that is a hallmark of prior–task mismatch.

The conservation-law prior shows a contrasting pattern. Because $\mathcal{L}^*_{\text{cons}} \approx 10^{-14}$, there is no irreducible floor. However, the conservation constraint is informationally weak: it constrains a single scalar invariant H on a 2-dimensional output. The resulting loss landscape contains spurious local minima along the conserved manifold, producing bimodal convergence rather than smooth degradation. The conservation λ sweep (Section 7.3) exhibits non-monotonic, seed-dependent behavior that cannot be characterized by the same dose–response framework as the midpoint prior.

B Conservation-Law Prior: Details

The Lotka–Volterra first integral is

$$H(u, v) = \delta u - \gamma \ln u + \beta v - \alpha \ln v.$$

The conservation loss penalizes $(\Delta H)^2 = (H(\hat{\boldsymbol{x}}_T) - H(\boldsymbol{x}_0))^2$. On ground-truth data, $|\Delta H|/|H| \approx 10^{-7}$, confirming the prior is exact to within numerical integration tolerances.

The conservation loss operates in physical (denormalized) space. Float32 normalization roundtrip introduces $\sim 10^{-7}$ maximum error, which is negligible. All training data satisfies strict positivity ($u > 0$, $v > 0$), ensuring the logarithmic terms in H are well-defined. No negative predictions were observed from untrained models within the tested initial-condition domain $[0.5, 2.5]^2$.

The bimodal optimization failure described in Section 7.3 is not caused by numerical issues in the loss computation, but by the geometric structure of the conservation constraint: $H(u, v) = c$ defines a closed curve in (u, v) space, and the conservation gradient is zero along this curve. This creates a degenerate direction in the combined loss landscape that becomes problematic as model capacity increases.

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